

Data Acquisition System
and Supervisory Control (SCADA)
Study is carried out at Khaperkheda Thermal Power

Station where the data acquisition system is in existence with DDC (Distributed Digital Control). Plant is divided in four areas for control and monitoring. There is 52 local buses and 4 intraplant buses and each intraplant bus has one redundant bus. When any intraplant bus becomes faulty, the redundant bus comes in service automatically. Basic diagram of the system is shown in fig. (1). In the figure intraplant buses with redundant bus is shown. Intraplant buses are coupled with local buses through bus coupler. Basically bus coupler is a micro controller connects an intra plant bus to any one local bus and it is commanded by local bus traffic director which is also a micro controller commanded by host computer to select a particular local bus whenever required.

Data are acquired in cycles continuously. When an address of particular module is sent by computer, output of that module floats on local buses and through intraplant bus it is acquired by host computer and displayed at CRT.

For control of any auxiliary, digital command is sent from control room which flows on intraplant and local bus and actuates the required module.

In fig (1) computer system is interfaced to the intraplant buses through communications port 1 2 3 & 4. Tap recorder is used for bulk memory storage. Control unit has its own software for control of different auxiliaries. commands is given to the control unit manually. Video control is automatic control. MDU (Magnetic disk unit) contains a software for data acquisition.

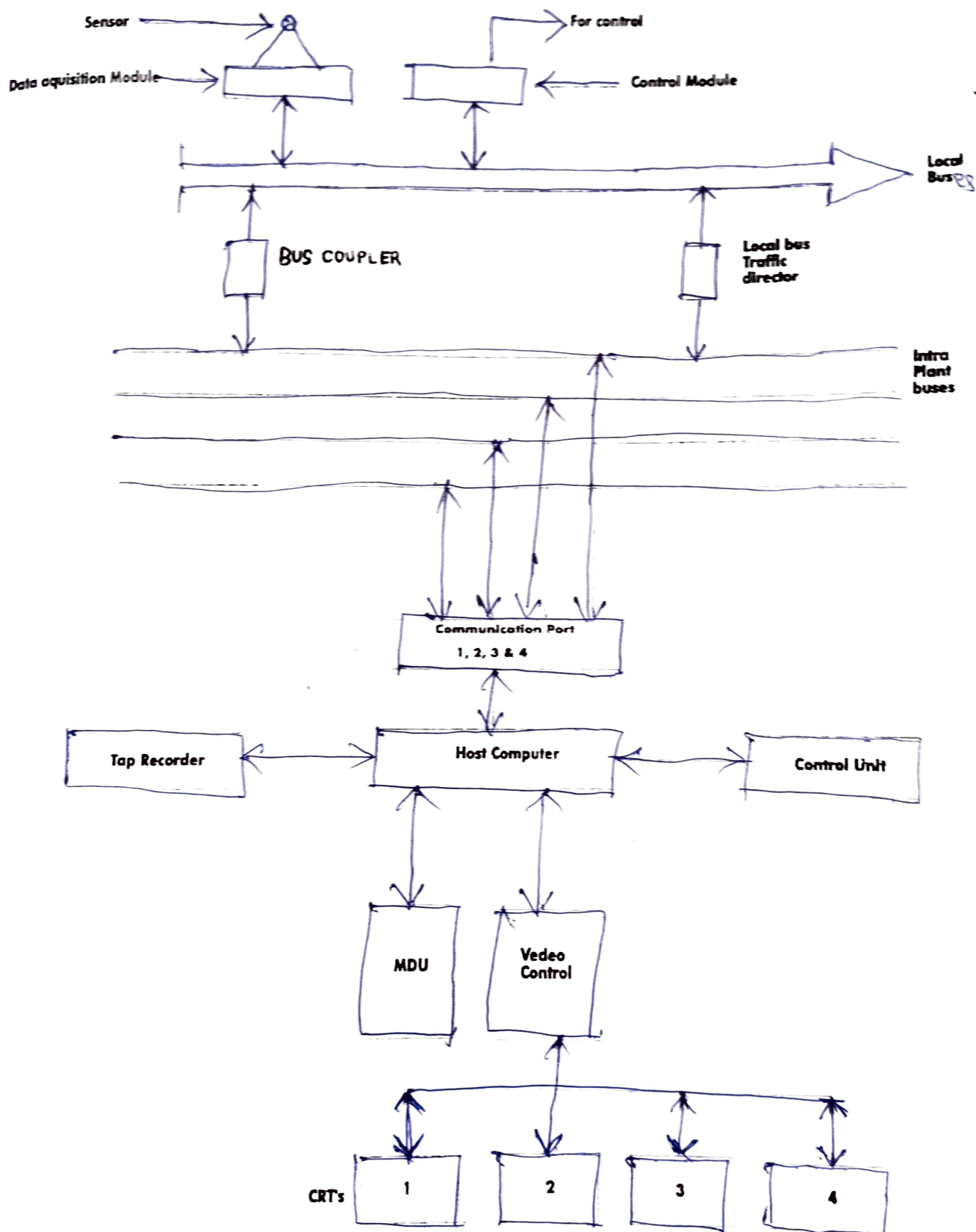


Fig (1)